

Welcome to Web Development

DAY 1

Today marks the beginning of your journey as a web developer. By the end of this session, you'll understand how the internet works, what HTML is, and you'll have built your very first webpage. Every professional developer started exactly where you are right now.

"Every website starts with simple HTML."



What is Web Development?

Web development means creating websites and web applications that run on the internet. Every site you visit - from Google to Instagram - was built by web developers writing code, designing layouts, and managing servers.

Designing Webpages

Creating the visual layout and structure users see on screen.

Writing Code

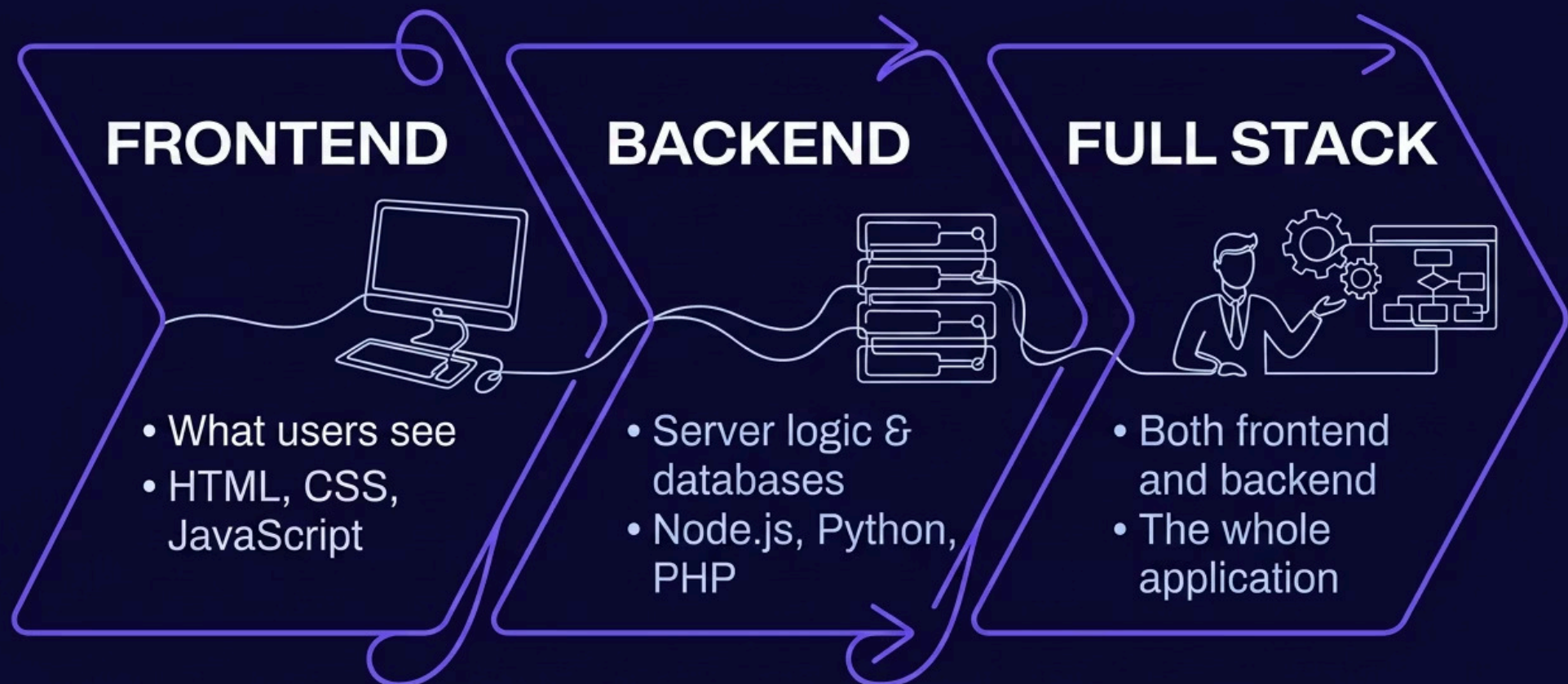
Using languages like HTML, CSS, and JavaScript to bring pages to life.

Adding Functionality

Making buttons, forms, and features work interactively.

Managing Servers

Storing data and serving pages to millions of users worldwide.



Frontend vs. Backend

Web development has two main sides that work together to create every website you've ever visited. Understanding the difference is one of the most important foundations of your learning journey.

Frontend Development

Everything the user can **see and interact with** in their browser. Frontend developers craft the visual experience.

- HTML - structure and content
- CSS - styling and layout
- JavaScript - interactivity

Examples: buttons, images, forms, navigation menus

Backend Development

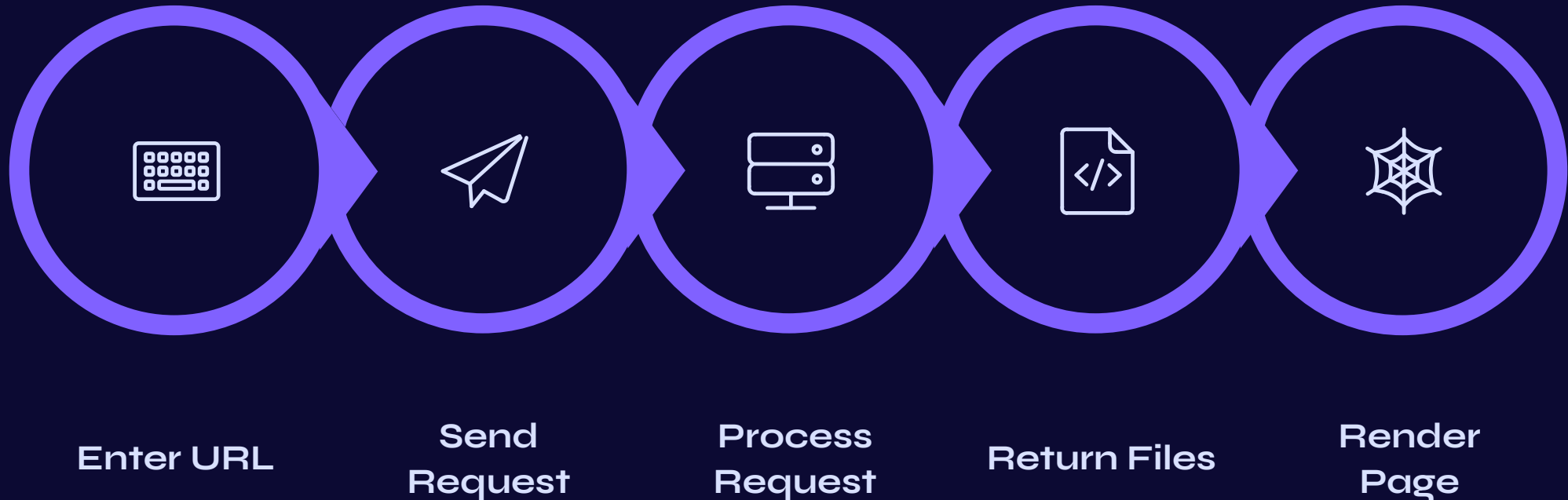
Everything that happens **behind the scenes** on the server. Users never see it, but it powers everything.

- Database handling and storage
- User authentication and login
- Server logic and APIs

Technologies: Node.js, Python, Java, PHP

How Websites Work

Ever wonder what actually happens when you type a URL and press Enter? It's a fascinating chain of events that takes place in milliseconds. Here's the step-by-step journey of a webpage request.



For example, when you open YouTube, your browser contacts YouTube's servers, which process your request and send back the video page files — all in under a second. Key terms to know: **Browser**, **Server**, **Domain**, **Hosting**, and **Internet**.

Introduction to HTML

HYPERTEXT MARKUP LANGUAGE

HTML is the foundational language of every webpage on the internet. It defines the **structure and content** of a page — not the style or behavior, just the raw building blocks. Think of it as the skeleton of a website.

What HTML Defines

- Headings and paragraphs
- Images and links
- Lists and tables
- Forms and inputs

Your First HTML Code

```
<h1>Hello World</h1>
<p>This is my first webpage.
</p>
```

These two lines create a heading and a paragraph — the building blocks of any page!



HTML Document Structure

Every valid HTML page follows the same basic structure. Think of it as a template — a blueprint that tells the browser exactly how to read and display your content. Let's break it down piece by piece.

The Full Structure

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First Page</title>
  </head>
  <body>
    <h1>Welcome</h1>
    <p>Hello Students</p>
  </body>
</html>
```

What Each Part Means

<!DOCTYPE html>

Declares the HTML version to the browser.

<html>

The root element that wraps everything.

<head>

Contains page info invisible to the user.

<title>

Sets the text shown in the browser tab.

<body>

All visible page content lives here.

Tags, Elements, and Attributes

These three concepts are the core vocabulary of HTML. Once you understand them, reading and writing any HTML code becomes much easier and more intuitive.

Tags

Keywords wrapped in angle brackets that tell the browser what type of content follows.

```
<p>This is a paragraph</p>
```

Elements

The complete structure — an opening tag, the content inside, and a closing tag working together.

```
<h1>My Title</h1>
```

Attributes

Extra information added inside the opening tag to modify or describe an element's behavior.

```

```

 Common attributes include: `src` (image source), `href` (link destination), `alt` (image description), `width`, and `height`. You'll use these every day!

Headings, Paragraphs, and Comments

These are the most fundamental HTML elements you'll use on every single page. Mastering them is the first real step in writing clean, readable HTML.

Headings — H1 to H6

HTML has six heading levels. Use `<h1>` for the main title, and smaller numbers for sub-sections.

```
<h1>Main Heading</h1>
<h2>Sub Heading</h2>
<h3>Section Title</h3>
```

Paragraphs

```
<p>This is a paragraph.</p>
```

Comments

```
<!-- This is a comment -->
```

Comments are completely ignored by the browser but are incredibly helpful for you and your teammates.

Why Comments Matter



Better Readability

Explain what a section of code does at a glance.



Easier Maintenance

Return to your code weeks later and still understand it.



Great for Teamwork

Help your teammates understand your code instantly.

Lists, Links, and Images

Three of the most-used HTML elements on any website. Together, they allow you to organize content, connect pages, and display visuals — the building blocks of almost every webpage you'll ever create.

Ordered List

Use when sequence matters — numbered automatically by the browser.

```
<ol>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ol>
```

Unordered List

Use for items without a specific order — shown as bullet points.

```
<ul>
  <li>JavaScript</li>
  <li>Python</li>
</ul>
```

Links

The `<a>` tag creates clickable hyperlinks to other pages or websites.

```
<a href="https://
linkedin.com">
  LinkedIn
</a>
```

Images

The `<img>` tag embeds images. Always include `alt` text for accessibility.

```

```

Practical Task & Homework

YOUR TURN!

It's time to put everything you've learned today into practice. Building something real — even something simple — is the fastest way to cement your new knowledge.

Today's Practical Task

Build a simple **personal webpage** using only HTML. Your page should include:

- Your name as the main heading
- A short introduction paragraph
- A list of your skills
- A profile image
- Links to your social media

Homework Assignments

Task 1: Create a complete "About Me" webpage with sections for About, Skills, and Contact.

Task 2: Practice using at least **20 different HTML tags**, including:

- `h1-h6`, `p`, `a`, `img`
- `ul`, `ol`, `li`, `br`, `hr`
- `div`, `span`, `table`, `form`, `input`, `button`

✔ Great work today! You've taken your first real step into web development. Keep experimenting — the best way to learn HTML is to write it every single day. 🚀